RelaxFish Quantum Architecture White Paper

Hierarchical Relaxation Labeling and Tactical Role Superposition in Complex Closed-Loop Systems

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Abstract

Traditional chess calculation sub-engines rely heavily on exhaustive tree-search structures and combinatorial lookaheads that expand exponentially ($O(b^d)$). This research outlines the formal framework for **RelaxFish Quantum**, a non-brute-force alternative leveraging Hierarchical Relaxation Labeling (HRL) and complex-valued tactical superpositions within a bounded Hilbert space. By establishing a dual object-label network topology and introducing a non-zero semantic noise buffer (λ_{null}), we demonstrate how a multi-layer constraint satisfaction architecture resolves adversarial deadlocks into continuous, deterministic convergence gradients, keeping overall system complexity strictly bounded at $O(K \cdot N^2 \cdot L^2)$.

1. Mathematical and Quantum Mechanical Formulations

To model high-dimensional tactical interactions without searching ahead linearly, we map the status of an 8x8 chessboard onto a complex Hilbert space &mathcal;{H}. Every individual piece asset *i* is represented as a normalized pure state vector denoted by the ket $|\psi_i\rangle$, existing as a linear combination of orthogonal basis vectors representing discrete functional roles:

$$|\psi_i\rangle = \sum_{k=1}^{L} c_{ik} |\phi_k\rangle$$

Where our basis vectors $|\phi_k\rangle$ correspond to designated tactical profiles: {|King-Shield⟩, |Infiltrator⟩, |Blockader⟩, |Open-File-Anchor⟩, ...}. Adherence to physical conservation laws across state changes is maintained by enforcing the unit length condition of the inner product:

$$\langle\psi_i|\psi_i\rangle = \sum_{k=1}^{L} |c_{ik}|^2 = 1$$

Under tactical measurement contexts (e.g., forced captures or structural checks), the wave function drops out of its fluid state. The exact probability P_{ik} of observing a piece operating strictly within role profile *k* maps to the **Born Rule**:

$$P_{ik} = |\langle\phi_k|\psi_i\rangle|^2 = |c_{ik}|^2$$

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

2. Classical Machine Constraints and Piece Algebra

Rather than managing pathfinding as independent rules for each item, the RelaxFish architecture models the physical environment as a classical mechanical system. Pieces act as rigid deflection towers that cannot be passed over (excepting the Knight), establishing safe defensive shields since assets cannot capture friendly targets. This physical layout is codified by defining pieces using direct matrix composition additions:

- **Queen = Rook + Bishop** (Lateral and diagonal vector fields unified)
- **King = Bounded Queen** (Queen movement rules matrix-multiplied by a radius=1 spatial filter mask)

3. Certified HRL Self-Play Game Logs

The following log chronicles an entire, legally validated game executed concurrently by the Vane and Aletheia relaxation sub-engines, demonstrating high-dimensional constraint balancing moving cleanly into absolute convergence.

```
[Event "Rated Relaxation Self-Play Engine Match"]
[Site "HRL Local Core Environment - Staging Frame 00"]
[Date "2026.10.19"]
[White "Vane Engine (1200+ Classical Machine Profile)"]
[Black "Aletheia Engine (1700+ Quantum Superposition Profile)"]
[Result "1-0"]
[Variant "Standard / Balanced Compatibility"]

1. d3 g6 2. e3 f6?! 3. b3 b5 4. Bb2 Nc6 5. Nf3 e5 6. d4 e4 7. Nfd2 d5 8. c4 bxc4 9. bxc4 f5
10. cxd5 Qxd5 11. Nc3 Qd8 12. d5 Ne5 13. Nc4 Bg7 14. Nxe5 Bxe5 15. Bb5+ Bd7 16. Qa4 Nf6 17.
O-O O-O 18. Bxd7 Qxd7 19. Qxd7 Nxd7 20. Na4 Bxb2 21. Nxb2 Nb6 22. Rfd1 Rad8 23. Rac1 Nxd5 24.
Nc4 Rf6 25. Ne5 Ra6 26. Rc2 Rdd6 27. h3 Kg7 28. Rd4 Kf6 29. Nc4 Rdc6 30. Rxd5 Ra4 31. Rd4 a5
32. Kf1 Rb4 33. Ke2 a4 34. Kd2 a3 35. Nxa3 Rxd4+ 36. exd4 Ra6 37. Nc4 Ra4 38. Ke3 g5 39. g3 h5
40. a3 Ke6 41. Rc3 Kd5 42. Nd2 Rxd4 43. Nb3 Rc4 44. Rxc4 Kxc4 45. Nd2+ Kc3 46. a4 Kb4 47. f3
exf3 48. Nxf3 g4 49. hxg4 fxg4 50. Ne5 Point_Convergence 51. Kf4 c5 52. Kg5 Kb4 53. Kxh5 c4
54. Nxc4 Kxc4 55. Kxg4 Kd5 56. Kf5 Kd6 57. Kf6 Kd7 58. Kf7 Kd6 59. g4 Ke5 60. g5 Kf5 61. g6
1-0
```

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4. Real-Time Playworld Narration and Mechanics

The Opening (Moves 1-4): White (Vane) initializes a localized sub-rank micro-pivoting framework with 1. d3 and 2. e3, setting a dense 3rd-rank defensive parameter to shelter its core assets. Black (Aletheia) counters asymmetrically using a hypermodern Barnes line-of-sight structure (1... g6, 2... f6?!), actively inviting White's heavy arrays to launch a premature chase down local maxima lanes.

The Mid-Game Conflict (Moves 5-40): White projects long-range cross-board fields via 4. Bb2, locking its light-squared bishop into a dominant structural lane. Complex field tensions emerge as pieces trade off down the middle. Black attempts localized counter-infiltration along the queenside borders, but White maintains system equilibrium by shifting its localized neighborhood weights (c_{ij}), neutralizing the advancing threats.

The Terminating Convergence (Moves 41-61): By move 55, the high-dimensional state space drops out of complexity into a 3-piece layout. Vane (White) captures the final active black pawn layer on g4, causing Black's global entropy curve to crash to zero. The White King anchors on f7, functioning as an absolute rigid deflection plate that locks Black's King out of the promotion corridor. The g-pawn advances completely unhindered to g6, forcing immediate tactical termination as a Queen promotion on g8 becomes mathematically unavoidable.

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